The University of Zambia

in association with

ZCAS University

BFS322 TREASURY MANAGEMENT

DECEMBER 2017

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Hanson Chemicals Agro group is a multinational company represented in five countries in Southern and Central Africa. The Board has been concerned with the financial performance of the subsidiary units, especially when it comes to the management of financial resources within the group. You have been recently appointed as the Group Chief Financial Officer and will be in charge of the treasury department. One of your key tasks is to reorganise the treasury function within the group in order to ensure better utilisation of financial resources.

(i) Discuss the role of the treasurer in the organisation and highlight the different functions that you will be expected to play in your new role. **(20 marks)**

(ii) What controls over the Treasury Department would you suggest the Board implements in order to ensure the effective and proper utilisation of resources in the Group? **(10 marks)**

(iii) You have recommended to the Board that in order to improve efficiency in the treasury function, the separate treasury's in the subsidiaries should be centralised under a single treasury function at the headquarters of the Group. Explain to management the benefits of this approach. **(10 marks)**

(Total: 40 marks)

SECTION B: **Attempt any THREE questions in this section**

Question 2

The treasurer is usually called upon to either manage a company's existing debt or procure new debt. In either case, this calls for knowledge of the broad variety of debt instruments available, as well as dealing with credit rating agencies.

(i) List and describe the features of at least five (5) such debt instruments available to the treasurer in Zambia. **(10 marks)**

(ii) As part of the process of issuing debt, the Treasurer has to deal with Credit Ratings Agencies. Highlight the criteria that these Credit Ratings Agencies base their analysis when rating a company's debt. **(10 marks)**

Question 3

One of the functions of the treasurer is to ensure that any surplus funds that are not needed in the company for either operating purposes of compensating bank balances are available for investment. The treasurer can use any number of investment vehicles to invest the surplus funds.

(i) Discuss the criteria that the treasurer will first consider before deciding on the different types of investments vehicles for investing surplus funds.

(6 marks)

(ii) Describe any four financial instruments available on the market that the treasurer can invest surplus funds.

(8 marks)

(iii) What kind of strategies can the treasurer use when investing the surplus funds (discuss any 3).

(6 marks)

Question 4

Treasury Bills		
Date: 08-11-2018 Tender No: 23/2018		
Tenure	Yield	Weighted Average Discount Rate
Tbs 91 Days (ZM3000005592)	14.4998	13.9939
TBs 182 Days (ZM3000006004)	15.0001	13.9562
Tbs 273 Days (ZM3000006012)	21.5000	18.5215
Tbs 364 Days (ZM3000006020)	23.0632	18.7506

(Source: www.boz.zm)

(i) Using the figures provided in the table above, draw the yield curve in Zambia and discuss the reasons why it is shaped the way it is. **(4 marks)**

(ii) The overall risks associated with lending increase over time. Discuss the various risks associated with lending over a long period of time. **(8 marks)**

(iii) In managing the interest rate risk that arises as a result of changes in the level of interest rates, discuss any two derivative instruments that the treasurer can use (additional marks will be earned if you use figures in your discussion). **(8 marks)**

Question 5

Exchange Rates of Zambian Kwacha against major foreign currencies:

Average Exchange rates: 14-11-2018		
Currency	**Buying**	**Selling**
USD	11.8922	11.9420
GBP	15.3980	15.4673
EUR	13.4168	13.4777
ZAR	0.8269	0.8314

(Source: www.boz.zm)

(i) From the figures provided above, state whether the kwacha is quoted directly or indirectly, giving reasons for your answer. **(2 marks)**

(ii) From the figures above, calculate the cross rate of exchange between ZAR and GBP. **(4 marks)**

(iii) Calculate the six-month forward USD/ZMK rate given the following information (use the mid-rate of USD/ZMK): **(8 marks)**

- Six-month Kwacha money market rate is 15%
- Six-month US dollar money market rate is 2.50%

(iv) Describe any three methods that the treasurer can use to hedge foreign exchange rate risk that do not involve the use of derivative instruments. **(6 marks)**

END OF EXAMINATION

MARKING KEY

Question 1

(i) Functions of treasury (2 marks for each)

- Treasury department is the financial centre of an organisation
- Key role is safeguarding and stewardship of organisation's financial assets and management of liabilities
- Responsible for implementing various financial decisions made by management and board

Specific roles of treasury:

1. Cash forecasting
   - Treasury compiles information on the receipts and disbursements of cash
   - Forecasts needed for investment purposes and to plan on investment vehicles and for cash scheduling purposes
2. Working capital management
   - Working capital involves changes in the levels of current assets and current liabilities
   - Treasurer needs to be aware of working capital levels and trends and advise management on impact of proposed policy changes
3. Cash management
   - Treasury staff uses information obtained from its cash forecasting and working capital management activities to ensure that sufficient cash is available for operational needs
4. Investment management
   - Treasury is responsible for the proper investment of excess funds.
   - Maximum return on investment of funds is not the primary concern,
   - rather it is not to put funds at risk and to match the maturity dates of investments with the company's projected cash needs
5. treasury risk management
   - interest rates that a company pays on its debt obligations may vary directly with market rates, which may present a problem if market rates are rising
   - a company's foreign exchange positions could also be at risk if exchange rates suddenly worsen
   - in both cases treasury staff can create risk management strategies and implement hedging tactics to mitigate the company's risk.
6. Management advice
   - Treasury staff monitors market conditions constantly and are able to advice management on the appropriate investment and borrowing decisions
7. Credit rating agency relationships

- When a company issues marketable debt, it is likely that a credit rating agency will review the company's financial condition and assign a credit rating to the debt
- The treasury liaises with the credit ratings agency and provides information

8. Banking relationships
    - The treasurer meets with representatives of any bank that the company uses to discuss the company's financial condition, fee structure, debt granted to the company etc
9. Fund raising
    - Treasury must maintain excellent relations with the investment community for fund raising purposes
10. Credit granting
    - The granting of credit to customers is one of the responsibilities of treasury
    - It allows treasury to have some control over the amount of working capital locked up in accounts receivables

(ii) Controls over treasury

1. Segregation of duties (4 marks)
    - Duties must be segregated among treasury staff, i.e. between front office and back office
    - E.g. trading activities must be separated from confirmation activities
    - Senior level treasury manager should approve all trades, another person should reconcile and account for all transactions
    - Someone outside the trading function should regularly compare brokerage fees or commissions to see if there are any unauthorised and unrecorded trades
2. Regular audits (2 marks)

- Internal audits should be scheduled in order to match actual transactions against company policies and procedures

3. Limits (4 marks)

- Treasurer should impose limit controls on a variety of transactions
- Limits can prohibit or severely restrict treasury staff from investing in certain types of financial instruments that present an unduly high risk of capital loss

(iii) Benefits if centralisation: (2 marks each)

- Transactions are handled in higher volumes by a smaller number of highly skilled staff
- Better management oversight
- Internal audit staff can review operations in a single location more easily

- Treasury activities involve complicated terminology that is incomprehensible to nontreasury specialists, it is better to centralise operations into a small well trained group

- Where an ERP system exists, all in house information required is available through the nearest computer terminal

Question 2

(i) Featurs of Debt Instruments: (2 marks for any five)

1. Commercial Paper

- Unsecured debt issued by a company and has a fixed maturity ranging from 1 to 120 days
- Used to meet short term working capital needs
- Sold at a discount from face value
- Company can sell directly to investors or through a dealer in an exchange for a commission

2. Field Warehousing Financing

- The finance company will segregate a portion of a company's warehouse
- All inventory within is collateral for a loan from the financing company
- The finance company will pay for raw materials as they are needed, and is paid back directly from accounts receivable as soon as the customer payments are received

3. Line of credit

- A commitment from a lender to pay a company whenever it needs cash, up to a pre-set maximum level
- Generally secured by company assets and interest charged is not far above the prime rate
- Bank charges an annual maintenance fee, irrespective of the amount drawn down on the loan

4. Asset Based loans

- Loan that uses fixed assets or inventory as collateral
- May also be financed on cash surrender value of a life policy, securities or real estate
- The bank will not lend more than 50% of the value of the raw materials and 80% of the value of the finished goods

5. Bonds

- A long term debt instrument issued with a fixed interest rate
- If the market interest rate is low, then investors will pay less for the face value of the bond, thereby driving up the net interest rate cost for the company and vice versa

- Rate of interest is inversely related to the price of the bond

6. Bridge loans

- Form of short term loan granted by a lending institution on the condition that the company will obtain longer term financing shortly that will pay off the bridge loan.
- Option commonly used when a company is seeking to replace a construction loan with a long term note that it expects to pay gradually over many years

7. Economic development loans

- Various state agencies have mandate to guarantee bank loans to organisations that need funds in geographic areas where it is perceived that social improvement goals can be attained e.g. Development Bank of Zambia and National Credit and Savings Bank (NATSAVE)

8. Long Term Loans

- Tend to be issued to companies that cannot issue bonds
- Require the company to grant lender senior status over all other creditors in event of liquidation
- Involves use of fixed loan repayments of principal and interest
- Company can lock in favourable interest rate for long time

9. Receivables Securitisation

- Achieves lowest interest rate available for debt
- Company creates special purpose entity (SPE) and transfers selection of receivables into the SPE.
- SPE sells the receivables to a bank conduit, which in turn pools receivables from multiple companies

(ii) Criteria used by credit rating agencies: (2 marks for each point)

- Rating process includes quantitative and qualitative analysis of the company
- Quantitative aspects focus on financial analysis and based on firm's financial reports
- They examine issuer's financials using proprietary models
- Qualitative side concerned with issues not related to the financials
- For corporate issuer may include analysing quality of management and the firm's competitiveness within its industry, as well as expected growth of industry and its vulnerability to business cycles, technology changes, regulatory changes and labor relations

Question 3

(i) Investment Criteria: (2 marks for each)

1. Safety of the principal being invested
   - When considering various forms of cash investment, the treasurer should first consider the safety of the principal being invested.
   - It would not do to invest the company's funds in a risky investment in order to earn very high returns if there is a chance that any portion of the principal will be lost.
2. Maturity and marketability
   - The treasurer must invest in securities that can easily be converted into cash should the need arise.
   - It is also best to invest in securities where a ready market exists for immediate resale.
3. Yield
   - This should be the final consideration after the previous factors have been considered.
   - Within the boundaries of appropriate levels of risk, maturity, and marketability, the treasurer can then settle for an investment with the highest yield.

(ii) Financial Instruments available for Investment purposes: (2 marks for any four)

a. Bankers acceptances

- Banks sometimes guarantee (or accept) corporate debt, usually when they issue a loan to a corporate customer, and then sell the debt to investors.
- Because of the bank guarantee, they are viewed as obligations of the bank

b. Bonds near maturity dates

- A corporate bond may not mature for many years, but one can purchase a bond that is close to its maturity date.
- Their tends to be a minimal risk of loss (or gain) on the principal amount of this investment, since there is low risk that interest rates will change so much in the short period of time left to maturity

c. Certificate of deposits

- Are term bank deposits, typically having durations of up to two years.
- Pay a fixed interest rate, though some variable rate CDs are available
- Perception that they are more secure than CP since they are issued by banks

- maturity date that will impact its value

d. Commercial Paper

- Large corporations issue short-term notes that carry higher yields than government debt issues
- There is an active secondary market for them, so they are very liquid
- CP is generally not secured, however investing in CP issued by 'blue chip' companies reduces default risk

e. Money market fund

- Package of government instruments, usually composed of treasury bills, notes, and bonds assembled by a fund management company
- Investment is highly liquid with many investors putting in funds for as little as a day
- There are varying levels of risk between different money market funds, since some funds are more active in trying to outperform the market (with an attendant increase in risk

f. Repurchase agreement (REPO)

- A package of securities that an investor buys from a financial institution, with an agreement that the institution will buy it back at a specified date
- Most commonly used for the overnight investment of excess cash
- Interest rate is equal to or less than money market rate

g. Treasury bills issuance

- Issued by the Central Bank on behalf of government to finance budget deficits
- Considered as risk free investments
- Issued at a discount to their face value
- Have lower yields than most other forms of investments

(iii) Investment strategies: (2 marks for each)

i. Earnings credit strategy

- At the most minimal level where the treasurer does nothing and leaves idle balances in the corporate bank accounts
- Earnings from idle balances offset its service fees
- Ideal for firms with minimal cash balances

ii. Matching strategy

- Treasurer matches the maturity date of an investment to the cash flow availability dates listed on the cash forecast.

iii. Laddering Strategy

- Involves creating a set of investments that have a series of consecutive maturity dates

iv. Tranched cash flow strategy

- Requires the treasurer to determine what cash is available for short, medium, and long term investment
- He can then adopt different investment criteria for each investment tranche e.g.:
  - Short term tranche is treated as cash that may be needed for operational requirements on a moments notice. Return on investment is not a key criterion-instead the treasurer focuses on
  - Medium term tranche- includes cash that may be required for use within the next 3 to 12 months and usually for predictable events such as periodic tax or dividend payments.
  - The long term tranche includes cash for which there is no planned operational use, and which the treasurer feels can be safely invested for at least one year.

Question 4

(i) Shape of the yield curve: (1 mark for drawing a yield curve and 3 marks for the explanation)

- The term structure of interest rates is often plotted on a yield curve
- A normal yield curve slopes upwards as borrowers who wish to borrow for long periods of time pay a higher rate of interest than short term borrowers
- The characteristic upward curve is attributable to liquidity preference and to risk
- The overall risks associated with lending increase over time, because other risks increase

(ii) Risks associated with lending long : (2 marks for each)

a. Default- either of interest or principal, which will generally increase with the length of the loan
b. Real capital loss- due to inflation, which increases with the length of the loan
c. Capital loss on marketable securities- which, if they have to be sold prior to maturity, increases with length of the loan
d. Changes in the lender's circumstances- which, making a particular loan within a portfolio inappropriate to new circumstances will increase with the length of the loan

(iii) Interest rate risk derivative instruments: (4 marks for any of two)

1. Interest rate swaps

- An interest rate swap is an exchange of interest rate commitments, such that a fixed rate commitment is exchanged for a floating rate commitment
- The parties to the swap retain their obligations to the original lenders
- The swap parties must accept counterparty risk

2. Forward Rate Agreement (FRA)

- A forward rate agreement (FRA) is an agreement whereby a firm can lock in an interest rate today for a period of time starting in the future
- On the future date, the two counterparties in the FRA settle up
- Depending on which way rates go, one will pay an amount of money to the other, representing the difference between the FRA rate and the actual rate

3. Interest Rate Futures

- Futures are standardised traded forms of FRAs
- FRAs are normally transacted with banks and other financial institutions and are tailor made to suit the dates and amounts that each individual company requires
- Interest rate futures are exchange traded and each contract is for a pre-specified amount and date

4. Interest Rate Options

- An option is a right but not the obligation, to carry out a transaction at a price set today, at some time in the future.
- Swaps, FRAs and futures are all contracts which two parties agree to transact and which must be carried out even if circumstances change
- An option, however, gives the buyer the choice of whether to transact or not
- A company would generally buy an option from an option seller or option writer
- An option is a form of insurance, and such, a premium is paid at the time the option is taken out, for the period of the option

Question 5

(i) Indirect or direct quotation: (2 marks)

- An indirect quote is where the US dollar is used as the base currency (i.e. = to 1) and while the domestic currency is the quoted currency or price (i.e. fluctuates).

- In a direct quotation, the US dollar would be the quoted currency while the domestic currency would be the base currency.
- Therefore in this case this is an indirect quotation as the Kwacha is the quoted currency while the dollar is the base currency.

(ii) Cross rate calculation for GBP/ZAR: (4 marks)

Bid	Offer
15.3989/0.8312	15.4673/0.8269
GBP/ZAR 18.5206	18.7052

(iii) Calculation of forward rate: (8 marks)

- The currency of the country having a higher interest rate trades at a discount i.e. add to the spot rate.
- The currency of the country having a lower interest rate trades at a premium i.e. deduct from the spot rate.
- (Premium)/discount = Exchange Rate x Interest Rate Differential x Days of contract/360
  (Premium)/discount = 11.9171 x (12-2.50/100) x 180/360
  = 0.7346 + 11.9171
- USD/ZMK 6 months forward rate = 12.6517

(iv) Foreign Exchange hedging strategies: (2 marks for each)

1. Accept the risk
   - Not hedging the exposure is the simplest strategy of all.
   - The company can accept the foreign exchange risk, and record the gains or losses on changes in the spot rate as they occur
2. Insist on home currency payment
   - Insist on being paid in the company's home currency, so that the foreign exchange risk shifts to your customer
3. Currency surcharge

- If a customer will not pay in a company's home currency, then bill the customer a currency surcharge if the company incurs a foreign exchange loss between the time of billing and payment.

4. Get paid on time
   - Insist on sales agreements with shorter payments terms, with sales staff being aggressive in collecting from foreign customers.
5. Foreign Currency Loans
   - The company should borrow in the foreign currency that matches the amount of the receivable.
   - When the customer pays off the receivable, the company uses the proceeds to pay off the loan.
6. Sourcing Changes
   - If there is a large amount of foreign currency cash flows coming from a specific country, then one way to hedge this risk is to start using suppliers located in the same country.
   - By doing so, the company can find a ready use for the incoming currency, by turning it around and sending it right back to the same country.
7. Foreign Currency Accounts
   - If a company regularly receives and pays out funds in a particular foreign currency, it may consider opening a foreign exchange account, in which It maintains a sufficient currency balance to meet its operational needs.